

Abstractive Summarization of Legal Bills: A Comparison of Transformer Architectures

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Course: CSE 676: Deep Learning, Spring 2026

1. Introduction

In the modern information age, the sheer volume of available data has made it increasingly difficult to process information efficiently. This issue is especially pronounced in the legal domain, where documents are often lengthy, densely written, and filled with specialized terminology. As a result, legal texts can be difficult to understand for those without formal training. Reading such material can impose a significant cognitive burden. Summarization offers a practical solution by condensing large amounts of information into shorter, more accessible forms. By removing less relevant details and highlighting key points, summaries can improve comprehension and retention. Importantly, the purpose of this approach is not to replace legal professionals, but to support them and to make legal information more accessible to a broader audience.

2. Dataset and Preprocessing

The primary objective of this project was to develop a system that improves the accessibility of legal documents. The focus was placed on abstractive summarization, in which the model generates new language to explain the original content, rather than simply extracting sentences directly from the source text. Abstractive models have the potential to restate complex ideas in clearer terms, making them more suitable for tasks like legal document summarization. The project evolved into a comparative study of three Transformer-based approaches:

1. A model built entirely from scratch
2. A standard pre-trained model
3. A hybrid model combining pre-training with custom modifications

This comparison allowed for an examination of the trade-offs between full architectural control and the advantages of large-scale pre-training.

The models were trained and evaluated using the BillSum dataset, which consists of United States Congressional and California state legislative bills paired with human-written summaries. To preprocess the data, the BART tokenizer was used across all experiments. This tokenizer operates at the subword level, enabling the models to handle rare or complex terms more effectively. Input texts were truncated to 512 tokens, and summaries to 128 tokens, creating a consistent sequence-to-sequence framework suitable for training.

3. Methodology: Model Designs and Implementation

Model 1: Custom Transformer (from scratch)

Following our checkpoint implementation, Model 1 is a vanilla Transformer built entirely using PyTorch primitives. This model consists of an encoder that processes the source bill text and a decoder that generates the summary. Since Transformers have no inherent notion of order, we implemented Sinusoidal Positional Encoding to inject sequence information:

$$P(pos, 2i) = \sin(pos / 10000^{2i/d})$$

$$P(pos, 2i+1) = \cos(pos / 10000^{2i/d})$$

The core of this model is the Multi-Head Attention (MHA) mechanism, which allows the model to attend to multiple parts of the bill simultaneously. We implemented scaled dot-product attention where the query (Q), key (K), and value (V) vectors are projected into 4 heads, concatenated, and passed through a final linear layer (Wo). To ensure stable training, we utilized Teacher Forcing and Causal Masking. The causal mask is a triangular boolean matrix that prevents the decoder from attending to future tokens, ensuring the model learns to predict the next word based solely on previous context. Finally, we implemented a manual Greedy Decoding loop with a repetition penalty of -10.0 to generate the summaries during inference.

Model 2: Standard Pre-trained BART Baseline

Model 2 utilizes the facebook/bart-base architecture, which acts as our baseline benchmark. Unlike Model 1, BART is a denoising autoencoder that has been pre-trained on massive datasets using five distinct noise functions: token masking, token deletion, text infilling, sentence permutation, and document rotation. This pre-training gives the model a deep intuition for English grammar and legal structure that a "from-scratch" model lacks.

Following the HuggingFace docs tutorials, we fine-tuned this model using the Seq2SeqTrainer and the AdamW optimizer. We set a learning rate of 2×10^{-5} and a weight decay of 0.01 to adapt the pre-trained weights to the specialized BillSum domain without causing catastrophic forgetting. For evaluation during training, we used `predict_with_generate=True`, allowing the trainer to calculate ROUGE scores at every epoch by running the full beam-search generation process (with `num_beams=4`). This model represents the "Industry Standard" approach, leveraging billions of parameters' worth of pre-trained knowledge.

Model 3: Hybrid BART Variant (Modified Architecture)

Model 3 serves as our "Researcher's Hybrid," designed to test how manual architectural changes affect pre-trained models. We wrapped the BART backbone in a custom `nn.Module` class and injected an extra Dropout layer ($p=0.2$) before the final vocabulary projection (the LM head). This was specifically intended to mitigate overfitting, as legal datasets are highly specialized and can cause a large model to "memorize" training samples rather than learning general summarization patterns.

In this model, we manually handled the training loop and loss calculation. We utilized Cross-Entropy Loss with *ignore_index = -100* to ensure that padding tokens were not penalized. Unlike Model 1, Model 3 uses BART's internal *shift_tokens_right* function to handle decoder inputs, ensuring that the target summaries are correctly aligned for next-token prediction. By combining the pre-trained power of BART with a custom regularization path, Model 3 explores whether minor structural modifications can improve the stability of domain-specific fine-tuning.

4. Deployment & UI

We developed a comprehensive web application using Streamlit with the aid of Google Gemini. This deployment focuses on providing an interactive comparison environment, allowing users to assess the real-world performance of our three distinct architectures side-by-side.

The application architecture consists of a three-tier design. The backend handles the loading of localized PyTorch weight files (specifically .pt files and the fine-tuned BART safetensors) into their respective nn.Module classes. The front-end interface features a dynamic sidebar for model selection and hyperparameter tuning. Users can adjust inference-specific variables such as Beam Search width and Max Length, providing control over the summary generation process. For Model 1, the UI executes our manual greedy decoding loop, while for Models 2 and 3, it leverages the Hugging Face generation API.

5. Results and Analysis

Dataset used: <https://huggingface.co/datasets/duccd/billsum/viewer/default/train?row=1>

Model 1: From-Scratch Transformer

Model Output

Investing in Rural School Lunch Act of 2003 - Amends the Child Nutrition Act to authorize the Secretary of Agriculture to make grants to eligible entities to provide technical assistance to eligible entities to provide technical assistance to eligible entities to provide technical assistance to eligible entities to provide technical assistance to eligible entities to provide technical assistance to eligible entities to provide child support services to children.

Inference Time

1.94 seconds

Model 2: Standard Fine-tuned BART

Model Output

Farm to School Improvements Act of 2010 - Amends the Richard B. Russell National School Lunch Act to direct the Secretary of Agriculture (USDA) to provide assistance through competitive matching grants and technical assistance to eligible entities for farm to school programs that: (1) improve access to local foods in schools and institutions participating in programs under the Act and the Child Nutrition Act of 1966 (CNA); (2) are designed to improve the nutritional health and well being of children; (3) procure healthy local foods from small and medium-sized farms for meals at eligible schools and institutions; (4) support experiential nutrition

Inference Time

5.25 seconds

Model 3: Hybrid BART + Dropout

Model Output

Farm to School Improvements Act of 2010 - Amends the Richard B. Russell National School Lunch Act to direct the Secretary of Agriculture to provide assistance, through competitive matching grants and technical assistance, to eligible entities for farm to school programs that: (1) improve access to local foods in schools and institutions participating in programs under this Act and section 4 of the Child Nutrition Act of 1966; (2) procure healthy local foods from small and medium-sized farms for meals at eligible schools and institutions; (3) support experiential nutrition education activities and curriculum planning that incorporates the participation of school children in farm and garden-based agricultural

Inference Time

5.04 seconds

Performance Benchmarks (ROUGE Scores)

	Model	ROUGE-1	ROUGE-2	ROUGE-L
0	Model 1	12.4000	3.1000	10.2000
1	Model 2	45.2000	22.5000	38.9000
2	Model 3	46.8000	23.1000	40.2000

6. Conclusion

This project demonstrates that while building Transformers from scratch provides good technical insight, pre-trained models like BART are significantly more effective for complex legal tasks. Specifically, model 3, which was a hybrid BART variant with dropout regularization, proved to have the best performance, being marginally better than even the standard pre-trained BART model (model 2).

7. Future Improvements

For Model 1, our output shows repetition, which is an issue. Future work could include replacing greedy decoding with beam search or top-k sampling because this reduces repetition and improves fluency. Since we were also limited by compute, we could increase the number of layers, heads, and embedding size so the model doesn't underfit language patterns. Future work could also include using 1024 tokens or long-sequence models (Longformer, BigBird). We could also implement some hyperparameter tuning such as tuning learning rate, batch size, dropout, number of epochs. For our experiment we used ROUGE scores only, we could add human evaluation (readability, coherence).

Team Contribution:

Team Member:	Project Parts:	Contribution %:
Radin Ahmad	Model 1, Model 2, Model 3, Report	50%
Shivam Joshi	Model 1, Model 2, Model 3, Report	50%

References

Vaswani, A., et al. (2017). "Attention Is All You Need."

Lewis, M., et al. (2019). "BART: Denoising Sequence-to-Sequence Pre-training."

Zhang, A., et al. (2023). "Dive into Deep Learning." (Sections 10.1 - 10.7).

<https://www.youtube.com/watch?v=Pk2BeaGbcTE>

Vocabulary : pg 336-341 d2l-en.pdf

Build function: pg 339 d2l-en.pdf

Padding and masking: pg 401 d2l-en.pdf

Embedding

<https://docs.pytorch.org/docs/stable/generated/torch.nn.Embedding.html>

```
self.embedding = nn.Embedding(vocab_size, embed_size)
```

pg 398

Attention mechanism and transformers pg 438 d2l-en.pdf

Multihead Attention pg 433 and 434 d2l-en.pdf

https://huggingface.co/docs/transformers/model_doc/bart

<https://dev.to/dm8ry/summarizing-text-using-hugging-faces-bart-model-14p5>

<https://huggingface.co/docs/transformers/tasks/summarization>

<https://arxiv.org/abs/1910.13461>

<https://discuss.huggingface.co/t/finetuning-bart-for-abstractive-text-summarisation/13430>